

Question	Answer	Marks
7(a)	$p = E / c$	C1
	$= (3.11 \times 10^{-19}) / (3.00 \times 10^8)$ $= 1.04 \times 10^{-27} \text{ N s}$	A1
7(b)(i)	$E = hf$ and $c = f\lambda$ so energy of one photon $= hc / \lambda$ $350 \times 10^{-3} = N \times (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (640 \times 10^{-9})$	C1
	$N = 1.1 \times 10^{18}$	A1
7(b)(ii)	$F = (\text{change in}) \text{ momentum} / \text{time}$	M1
	Clear use of $p = E / c$ and $t = E / P$ to complete the algebra and arrive at the final equation: e.g. $F = [E / c] / [E / P] = P / c$	A1
7(c)(i)	maximum wavelength (of electromagnetic radiation) that causes electrons to be emitted (from surface of metal)	B1
7(c)(ii)	work function $= 2.26 \times 1.60 \times 10^{-19} \text{ (J)}$ $E = hc / \lambda$ so $(2.26 \times 1.60 \times 10^{-19}) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda_0$	C1
	$\lambda_0 = 5.50 \times 10^{-7} \text{ m}$	A1

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